

Resampling methods

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Resampling methods

1. Draw samples from a training set.
2. Fit the model to each sample.

Aims can be:

- ▶ Estimate the variability in the results of the fit.
 - ▶ Estimate the test error of a model.
 - ▶ Select the flexibility of the model to fit.
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- Cross-Validation
 - Bootstrap

Training set and test set

For prediction problems one wants a small **test error rate**.
A small error rate on the training data does not imply a small error rate on the test data.

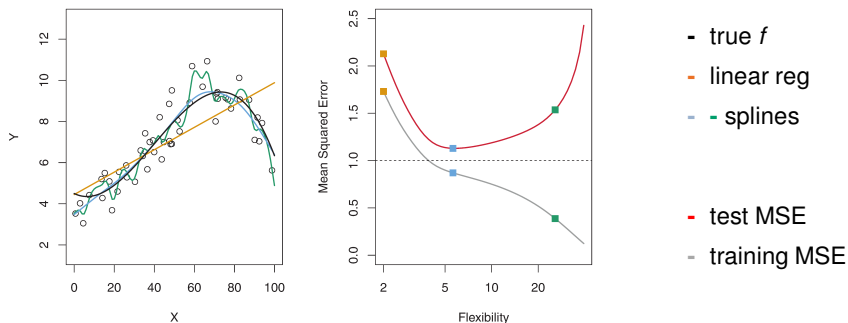


Figure adapted from James, Witten, Hastie, Tibshirani

- ▶ But what to do if there is no test data set?
- Define test set by holding out part of the data.

Validation set

Divide the data into two data sets:

- ▶ Training set (blue): used to fit the model
- ▶ Validation set (orange): used to assess the model



Figure from James, Witten, Hastie, Tibshirani

Drawbacks with validation set

Example: Linear or polynomial regression

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \dots$$

Validation error estimates for 10 different splits of the data into training and validation sets.

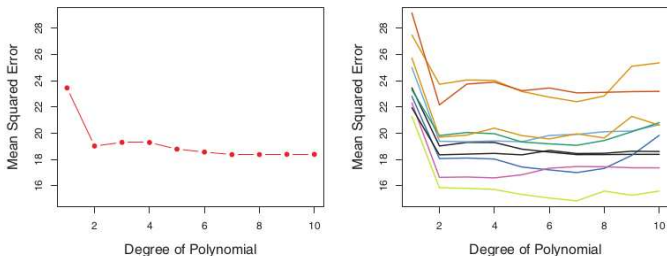


Figure from James, Witten, Hastie, Tibshirani

- ▶ The test error rate depends on the division of the data.
- ▶ Models are better with more training data, but a subset of data is hold out.

Leave-one-out cross-validation

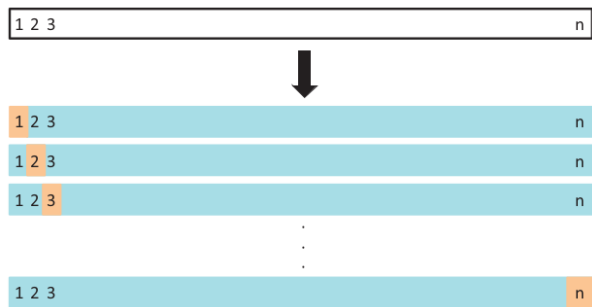


Figure from James, Witten, Hastie, Tibshirani

- ▶ One observation (Y, X) is left out and $n - 1$ are used for training.
- ▶ A prediction is made for the left out data.
- ▶ The process is repeated n times, leaving one different observation out each time.
- ▶ The error rate of the model is estimated as the average error rate for the n predictions.

k -fold cross-validation

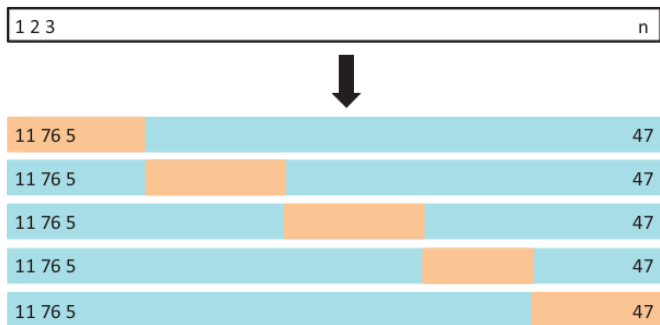


Figure from James, Witten, Hastie, Tibshirani

- ▶ The data set is randomly divided into k groups (folds) of similar size.
- ▶ One fold is left out and a model is trained on the rest of the data.
- ▶ A prediction is made for the left out fold.
- ▶ The process is done k times, leaving out a different fold each time.

Cross validation

Example: Linear or polynomial regression

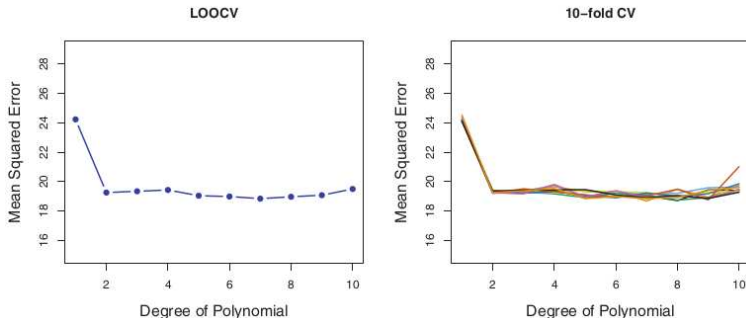


Figure from James, Witten, Hastie, Tibshirani

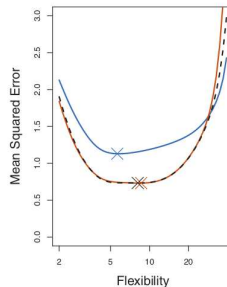
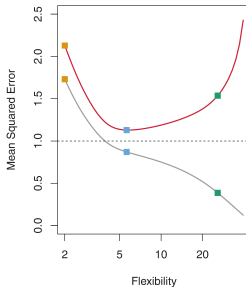
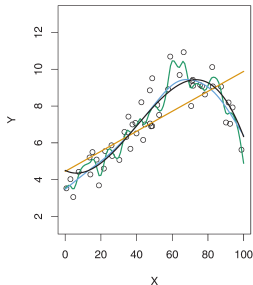
- ▶ LOOCV is a special case of k -fold CV ($k = n$)
- ▶ LOOCV produces just one estimate of the error rate (no randomness).
- ▶ k -CV results in variability of the estimate of the error rate.
- ▶ LOOCV is computationally expensive when n is large.

Cross validation

Aims of cross -validation:

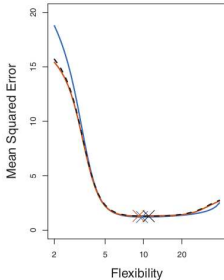
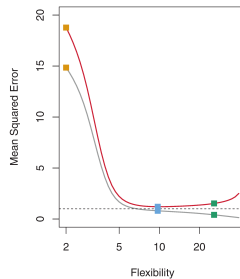
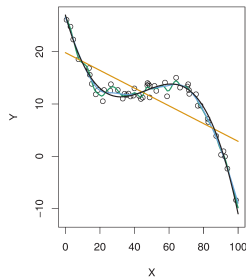
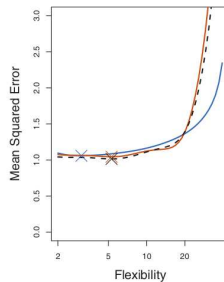
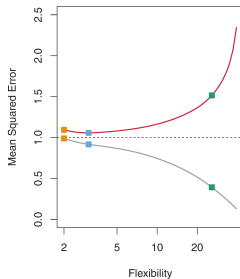
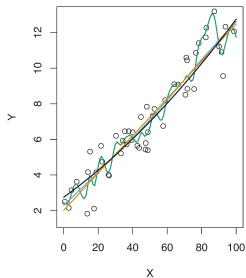
- ▶ Estimate how accurate is a given model - the actual estimated MSE is relevant.
- ▶ Identify what statistical learning method is best for a given data - the minimum point in the estimated MSE curve is relevant.

Example with simulated data:



blue: true MSE; orange: 10-fold; dashed: leave-one-out

Cross validation



blue: true MSE; orange: 10-fold; dashed: leave-one-out

Bias-variance trade-off for k-fold cross-validation

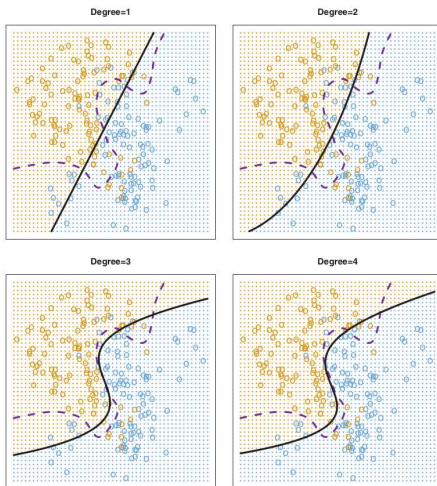
What k to choose for cross-validation?

- ▶ Dividing the data in 2 $\rightarrow$ half the data is used to train $\rightarrow$ overestimation of the test error (bias).
 - ▶ Large $k \rightarrow$ small bias.
- ▶ Large $k \rightarrow$ few observations in each test data set $\rightarrow$ high variance of the test error estimate.
 - ▶ Large $k \rightarrow$ high variance.
- ▶ $k = 5$ or 10 are commonly used.

Cross-validation for classification problems

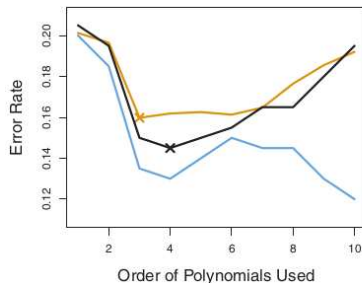
Simulated 2 dimensional data, logistic regression fits:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2 + \dots$$

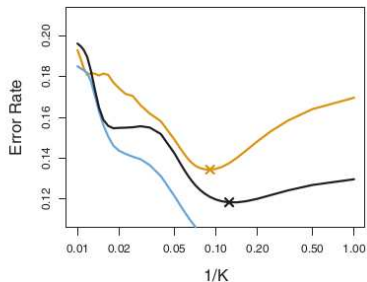


Cross-validation for classification problems

Logistic regression



K-nearest neighbors classifier



brown: test error; blue: training error; black: 10-fold CV error

Figure from James, Witten, Hastie, Tibshirani

Bootstrap

Aim: Quantify the uncertainty associated with an estimator or method.

Assume one wants to estimate

- ▶ α , which is a function of the observations;
- ▶ the standard deviation of the estimator.

Procedure:

- ▶ From 100 observations obtain the estimate $\hat{\alpha}_1$
- ▶ Repeat 1000 times to obtain the estimates $\hat{\alpha}_1, \hat{\alpha}_1, \dots, \hat{\alpha}_{1000}$
- ▶ Calculate the mean and standard deviation of the $\hat{\alpha}$ s.

Samples

Examples of the 100 pairs of observations (X, Y)

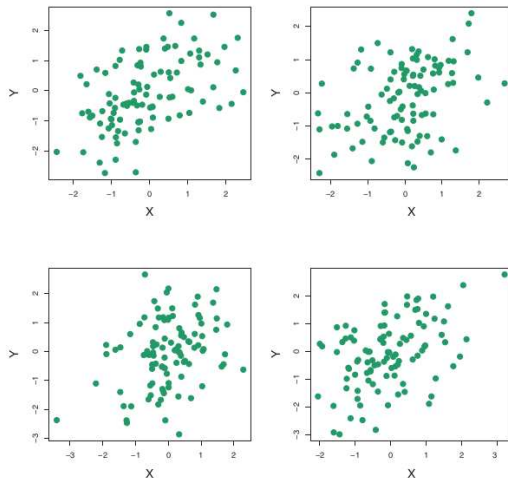
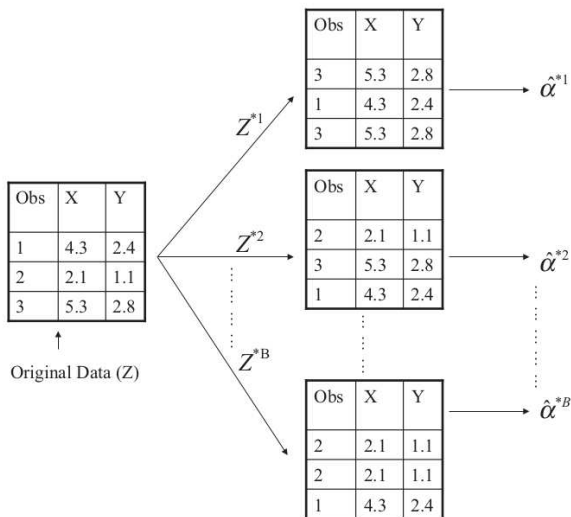


Figure from James, Witten, Hastie, Tibshirani

Creating the bootstrap samples

- ▶ Usually it is not possible to obtain many sets of the data.
- ▶ New sets can be created by sampling with replacement.



Bootstrap estimation

Example: estimation of α with 1000 data sets and 1000 bootstrap samples.

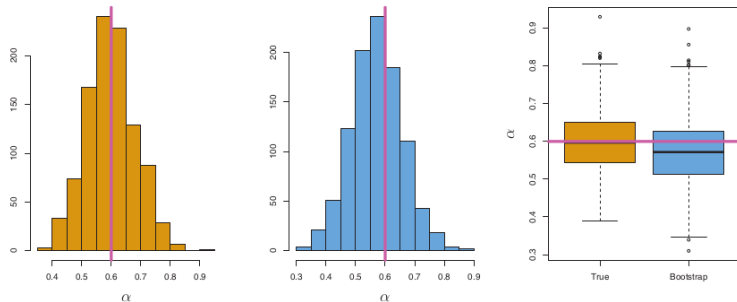


Figure from James, Witten, Hastie, Tibshirani

Standard error of the bootstrap estimates:

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B \left(\hat{\alpha}^{*r} - \frac{1}{B} \sum_{r'=1}^B \hat{\alpha}^{*r'} \right)^2}$$